

# Proactive Memory for Event-Driven LLM Agents

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## Abstract

Proactive LLM agents anticipate service opportunities from ongoing observations and accumulated history. A central challenge is determining which historical observations remain applicable to the current event and can support a timely service decision. We propose Proactive Memory (ProactMem), combining bounded recent events, source-checked daily trajectories, and replay-assessed temporal patterns. Daily validation checks extracted facts against source events, while historical replay assesses recurring patterns before they are matched to current events. We also construct ProactiveStream, a longitudinal benchmark covering three application scopes, where agents process isolated event streams chronologically under episode-aware opportunity labels. ProactMem improves Effectiveness by 4.3–15.7 percentage points over the strongest external baseline in each scope. Ablation studies support the contributions of hierarchical memory organization and validation to service-opportunity detection.

## 1 Introduction

Proactive LLM agents anticipate useful assistance as a user’s situation evolves, taking initiative before an explicit request. Recent systems study when to propose a task, detect emerging needs, and use contextual or historical information to guide that judgment (Lu et al., 2025; Yang et al., 2025; Xie et al., 2026; Li et al., 2026). Across our Sports, Home, and Code scopes, this requires interpreting a continuing stream of observations and deciding whether a service opportunity warrants attention now.

Historical context is central to this decision. Figure 1 illustrates how earlier evening walks can inform a service judgment when current activity is low and the user is available. Daily trajectories and temporal patterns retain this context beyond a

recent-event window. The agent must connect that history to a concrete, timely service.

Agent memory systems retain observations, consolidate experience, and retrieve related information (Park et al., 2023; Packer et al., 2023; Chhikara et al., 2025; Xu et al., 2025). To turn this history into decision context, proactive agents must assess whether an abstraction reflects its source events, whether a proposed temporal pattern has recurring support, and whether that pattern applies now. Our central question is *how to turn longitudinal history into reliable, currently applicable context for proactive service decisions*.

We propose Proactive Memory (ProactMem; Figure 2), a hierarchical memory framework that organizes event histories into source-checked daily trajectories and replay-assessed temporal patterns for proactive service decisions. Bounded Event Memory preserves recent observations, while Daily Trajectory Memory retains source-checked abstractions beyond the recent window. Temporal Pattern Memory consolidates recurring conditions from these daily trajectories and assesses their historical support through replay. At decision time, the agent combines recent events and daily trajectories with currently applicable patterns. The decision LLM assesses whether current observations or historical context support a concrete, timely, low-risk service; episode control tracks whether the proposed opportunity has already been covered.

We construct *ProactiveStream* to study these decisions over 4,539 events from 17 isolated streams. Its three scopes preserve causal history and define episode-aware reference opportunities, allowing evaluation of memory formation and repeated service decisions within the same streams. ProactMem improves Effectiveness by 4.3–15.7 percentage points over the strongest of five external baselines in each scope. Targeted ablations examine the roles of hierarchical memory organization and support checks.

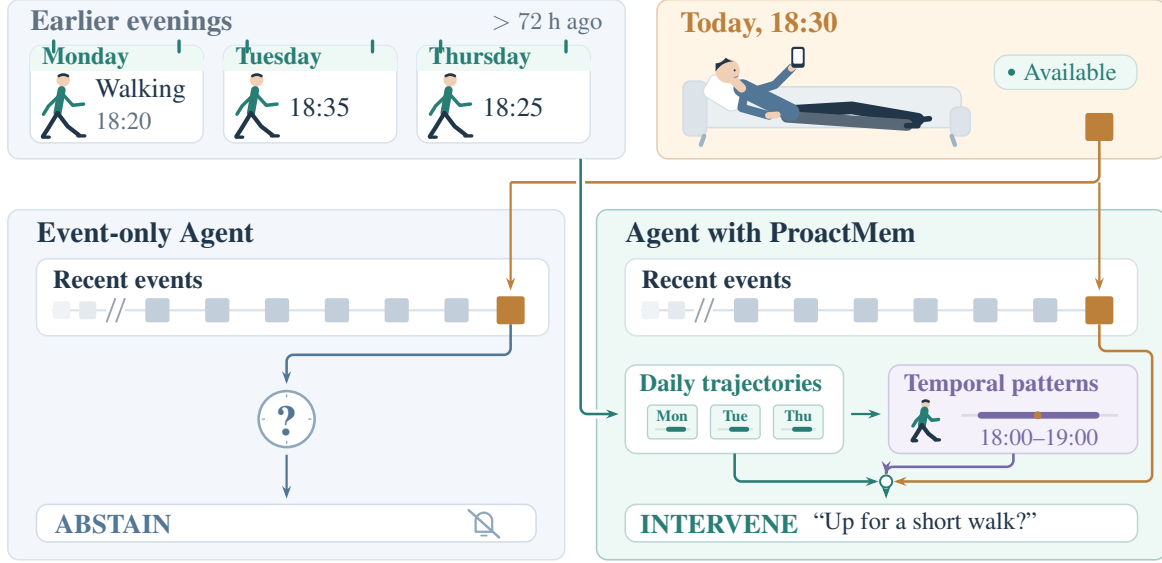


Figure 1: Illustrative Sports scenario. Both agents receive the same current event and recent history. ProactMem retains earlier walking observations as daily trajectories and temporal patterns, providing context for a timely activity suggestion. Responses depict possible decisions.

Our contributions are:

- **Method.** ProactMem combines source-checked daily abstraction, replay-based pattern assessment, and selection of currently applicable patterns to supply historical context for proactive service decisions.
- **Benchmark.** ProactiveStream supports longitudinal evaluation of memory formation and repeated service decisions, preserving causal history, entity isolation, and episode-aware reference opportunities across three scopes.
- **Empirical findings.** ProactMem achieves higher event-level Effectiveness and Balance than the evaluated external baselines in all three scopes; targeted ablations support the contributions of hierarchical organization and validation.

## 2 Problem Formulation

Let  $u$  denote an independently evaluated entity, such as a participant, household, or repository, with a time-ordered public event stream

$$E^u = (e_1^u, \dots, e_{T_u}^u), \quad e_t^u = (\tau_t^u, x_t^u), \quad (1)$$

where  $T_u$  is the number of events for entity  $u$ ,  $\tau_t^u$  is the event time, and  $x_t^u$  contains the scope-specific public fields observable at that time. We omit

the entity superscript  $u$  when it is unambiguous. Events follow a fixed chronological order with deterministic tie-breaking. Before processing  $e_t^u$ , an agent has a causal state  $Z_t^u$  constructed only from events and finalized outputs preceding  $e_t^u$  in this order. It produces

$$(\hat{y}_t, \hat{m}_t) = \pi(e_t^u, Z_t^u), \quad \hat{y}_t \in \{0, 1\}, \quad (2)$$

where  $\hat{y}_t = 1$  denotes INTERVENE,  $\hat{y}_t = 0$  denotes ABSTAIN, and  $\hat{m}_t$  is an optional evidence-grounded service message. Here  $\pi$  denotes a general inference policy over the current event and causal state;  $\hat{m}_t$  may be empty under ABSTAIN. After finalizing the decision and optional message, the agent updates its state through an update function  $U$ :

$$Z_{t+1}^u = U(Z_t^u, e_t^u, \hat{y}_t, \hat{m}_t). \quad (3)$$

Each entity maintains isolated state, and  $\pi$  accesses only the current public event and causally available same-entity information.

The objective is to identify a timely, actionable *service opportunity* supported by the current event or prior same-entity observations. Let  $y_t \in \{0, 1\}$  denote the frozen reference decision for  $e_t$  under the annotation protocol in Section 4.2; all evaluated policies share these fixed references. Labels and annotation metadata remain evaluator-private. Events with either reference label share the same scope-specific public input schema.

### 3 Proactive Memory (ProactMem)

ProactMem uses three complementary memory layers (Figure 2): Event Memory for recent public events, Daily Trajectory Memory for source-checked daily abstractions, and Temporal Pattern Memory for replay-assessed recurring patterns. Construction checks source fidelity and historical support, retrieval selects currently applicable patterns, and the decision LLM assesses present service value.

#### 3.1 Memory Construction and Update

**Bounded Event Memory.** For event  $e_t$ , Event Memory retains prior events in their original public representation, intersecting a temporal horizon  $\Delta$  with capacity  $K$ :

$$S_t = \text{Tail}_{K-1}((e_i)_{i < t, 0 \leq \tau_t - \tau_i \leq \Delta}). \quad (4)$$

Here  $\text{Tail}_{K-1}$  selects the most recent  $K-1$  events in the filtered sequence. We use  $\Delta = 72$  hours and  $K = 100$ : at most 99 prior events accompany the separately supplied current event. An independent daily buffer retains all committed events of the open local day, preserving complete input for daily construction.

**Fact-validated Daily Trajectory Memory.** After a local day closes, a construction LLM summarizes its buffered events as trajectory segments with intervals, typed facts, and event citations. Let  $F(\cdot)$  denote normalized checkable facts obtained from typed fields. For segment  $d$  and its cited events  $\mathcal{E}(d)$ , source consistency requires

$$F(d) \subseteq \bigcup_{e \in \mathcal{E}(d)} F(e), \quad \mathcal{E}(d) \subseteq E_{\text{completed day}}^u. \quad (5)$$

Here  $E_{\text{completed day}}^u$  contains the same entity’s events from that completed day. The validator checks entity, interval, citations, and canonicalized scalar values at compatible field paths. Acceptance requires source-supported typed facts and resolved citations. Accepted daily trajectories form  $W_t$ , retaining historical context beyond Event Memory. Appendix B details field handling and buffering.

**Replay-assessed Temporal Pattern Memory.** Weekly consolidation groups seven completed daily records per entity in time order, with each record assigned to one cycle. A cycle may span more than seven calendar days. An LLM proposes

a trigger  $g$ , expected pattern  $p$ , time window, optional weekday restrictions, and source daily identifiers. Replay checks these conditions against validated daily facts: an observable triggered day is fulfilled when  $p$  occurs at or after the trigger within the window and violated otherwise. Missing observations are censored, inapplicable days are excluded, and each date contributes at most once. Let  $n_f$  count fulfilled (supporting) days and  $n_v$  count violated days:

$$\text{conf}_{\text{replay}} = \frac{n_f}{n_f + n_v}, \quad \text{lift} = \frac{\text{conf}_{\text{replay}}}{\text{base}(p)}. \quad (6)$$

The base rate is the frequency of  $p$  over observable days under the same window/weekday restrictions across both triggered and other observable days. Zero denominators yield zero confidence or lift, respectively.

A well-formed proposal becomes *Active* when  $n_f \geq 3$ , confidence  $\geq 0.5$ , and lift  $\geq 1.0$ ; otherwise it remains decision-invisible *Candidate*. Malformed proposals are rejected. Consolidation resets the online daily view, retaining records for audit. Later observations update pattern confidence and lifecycle: repeated violations can lower retrieval priority or suspend eligibility. Appendix B.2 specifies these transitions. Historical support characterizes observed recurrence; future applicability and present service value require separate assessment.

#### 3.2 Memory Retrieval and Selection

Event Memory supplies the bounded sequence and  $W_t$  supplies all completed daily trajectories in the current cycle. Pattern retrieval additionally checks entity-local trigger context, weekdays, and temporal applicability with a 30-minute lead/grace interval. Matching considers current and previously recorded causal trigger occurrences. An occurrence is fulfilled when its expected behavior has been observed; censored occurrences have insufficient observations for assessment. Selection excludes both states. For pattern store  $L_t^u$ , lifecycle eligibility  $\text{Eligible}_t$ , trigger/time matching  $\text{Match}_t$ , and occurrence state  $s_t$ ,

$$R_t = \text{Top3} \{ \ell \in L_t^u : \text{Eligible}_t(\ell) \wedge \text{Match}_t(\ell), s_t(\ell) \notin \{\text{fulfilled}, \text{censored}\} \}. \quad (7)$$

The top-three selection ranks eligible patterns by occurrence state, lifecycle, confidence, and temporal distance (Appendix B.2). All matching uses

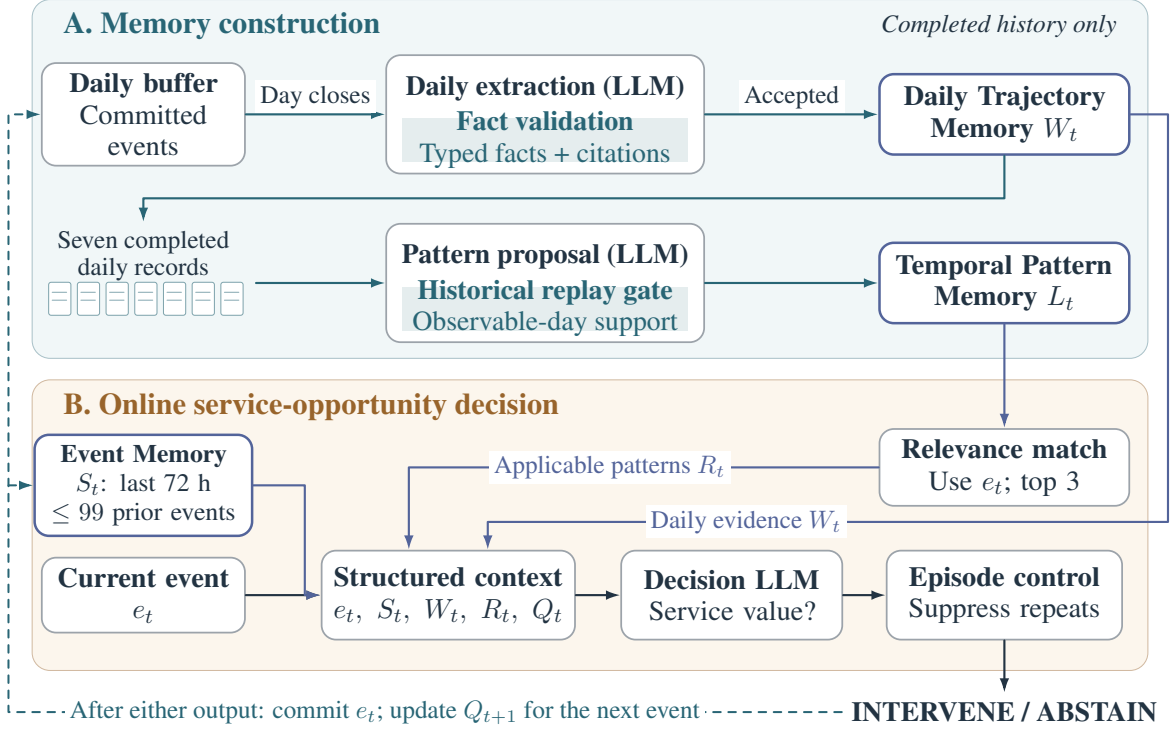


Figure 2: ProactMem separates memory construction (A) from online decision-making (B). Daily extraction is checked against cited events; temporal proposals are assessed by replay over completed daily records. Only lifecycle-eligible, currently relevant patterns enter the structured context. The decision LLM assesses service value, and the shared controller filters proposed interventions to suppress repetition. Dashed feedback commits the current event after the output is finalized, whether INTERVENE or ABSTAIN;  $Q_t$  describes observable episode state.

causally available observations to assess present applicability.

### 3.3 Proactive Decision

The decision context combines the current event, recent events, daily trajectories, selected patterns, and observable episode state:

$$C_t = (e_t, S_t, W_t, R_t, Q_t), \quad (8)$$

where  $Q_t$  describes the service opportunity’s observable state; the controller separately records previously allowed interventions to track service coverage. The LLM produces  $(\tilde{y}_t, \tilde{m}_t) = \pi_\theta(C_t)$ , proposing intervention when the current event *or* historical context supports concrete, timely, feasible, low-risk assistance, and abstention otherwise. Pattern matching supplies evidence, not an intervention command: the LLM can abstain even when a historical pattern applies.

Within each scope, all methods apply the same Episode Controller policy to their own context and intervention history. The controller compares proposed services with previously allowed interventions and converts duplicates to ABSTAIN. Scope-specific changes in opportunity state, service iden-

tity, contextual patterns, or local day permit renewed assessment; each intervention originates from an LLM proposal. The controller finalizes the proposal as  $(\hat{y}_t, \hat{m}_t)$ , the output defined in Section 2. Appendix B.3 specifies matching fields, phase transitions, and re-entry rules.

Inference prepares prior memory, stages  $e_t$ , selects relevant context, obtains and finalizes the decision, and then commits  $e_t$ . Maintenance incorporating  $e_t$  can affect only later decisions. Reference labels remain evaluator-private throughout construction and retrieval.

## 4 ProactiveStream Benchmark

ProactiveStream evaluates service-opportunity detection in longitudinal event streams, where agents assess incoming observations using their own accumulated history. It comprises 4,539 events across 17 isolated streams in three service scopes, with 729 reference interventions (Table 1).

### 4.1 Scopes and Public Events

**Sports** uses participant activity and availability contexts for optional walking and everyday physical-activity assistance.

Table 1: Composition and provenance of ProactiveStream across Sports, Home, and Code. Each stream is an independent memory and replay unit. Reference interventions denote service opportunities under the annotation protocol.

| Scope  | Primary source | Entity          | Streams | Input events | Temporal coverage                 | Reference interventions |
|--------|----------------|-----------------|---------|--------------|-----------------------------------|-------------------------|
| Sports | HeartSteps     | participant     | 9       | 1,456        | 367 observed participant-days     | 365                     |
| Home   | CASAS          | home            | 4       | 1,624        | 14 consecutive days/home          | 190                     |
| Code   | GH Archive     | repository      | 4       | 1,459        | longitudinal repository histories | 174                     |
| Total  | three scopes   | isolated stream | 17      | 4,539        | three longitudinal settings       | 729                     |

**Home** uses time-bounded home-sensor observations for gentle, low-risk check-ins. Each sensor segment becomes available at interval closure.

**Code** uses repository activity at GitHub’s native event granularity for low-risk triage, review, and maintenance, with one decision per event.

Each participant, home, or repository defines an isolated chronological stream, allowing earlier observations to inform later service opportunities. Public inputs contain contemporaneously available observations, using the same scope-specific schema for both label classes. Agents construct their own memory from these events; annotation and scoring metadata remain evaluator-private. Appendix A details source datasets, fields, and preprocessing.

## 4.2 Reference Labels and Causal Replay

Annotators assess  $e_t$  and its complete prior same-stream history  $\mathcal{H}_{<t}$ . Let  $A$  indicate a concrete, timely, feasible, low-risk opportunity supported by  $e_t$ ,  $B$  an opportunity at  $e_t$  supported by historical observations, and  $N_t^{\text{ref}}$  reference-episode novelty:

$$y_t = 1 \iff (A(e_t) = 1 \vee B(e_t, \mathcal{H}_{<t}) = 1) \wedge N_t^{\text{ref}} = 1. \quad (9)$$

The reference protocol labels the first reliable actionable window INTERVENE and covered windows ABSTAIN, allowing renewed opportunities when support strengthens, the service changes, or an episode reopens. Reference novelty follows the reference opportunity sequence independently of model outputs and is shared across methods. The event-level labels therefore assess timely identification of service windows defined by the annotation protocol.

LLM-assisted screening prioritizes candidate opportunities; three team members then independently label every event while blinded to evaluated-

system outputs, followed by cross-review and adjudication. For a separate-team review, a fixed SHA256 rule selects one stream and 150 contiguous targets per scope from public metadata, requiring at least seven days of prior history and retaining each complete causal prefix. Three reviewers label these 450 targets with model outputs hidden and form consensus references. Section 5.2 reports reference-sensitivity scoring of fixed diagnostic outputs. Main comparisons use protocol-fixed evaluation on the jointly developed benchmark, with labels, prompts, memory bounds, and validation settings frozen before the corresponding runs.

Each stream is replayed chronologically from empty method state, with access to the current event and strictly prior same-stream context. Each decision is finalized before the current event is committed to memory; reference labels are joined after inference for scoring. Scoring includes the initial events and continues as history accumulates.

## 5 Experimental Evaluation

We examine overall effectiveness (RQ1), memory organization (RQ2), and validation mechanisms (RQ3), with supplementary checks of reference sensitivity, episode behavior, and model choice.

### 5.1 Experimental Settings

**Baselines.** We evaluate five external methods:

- **ProactiveAgent** uses its released prompt and proposal/null contract, with causal event–decision history capped at 1,000 events (Lu et al., 2025; THUNLP, 2026).
- **ContextAgent** adapts in-context proactive-service judgment to public events (Yang et al., 2025).
- **Mem0** extracts, consolidates, and retrieves same-entity memories (Chhikara et al., 2025; Mem0, 2026).

- **A-MEM** constructs linked notes, updates associations, and retrieves relevant notes (Xu et al., 2025).
- **Graphiti** stores event relations in a temporal graph and retrieves same-stream facts (Zep, 2026).

**Implementation Details.** The primary model is gpt-5.6-luna, with temperature zero and thinking disabled; ProactMem uses the official OpenAI API. Methods share task semantics and Episode Controller, with interface-specific prompts and budgets. Event-only, Event+Daily, and ProactMem share their decision prompt, schema, and 72-hour/100-event bound including the current event. A-MEM and Graphiti augment that bound with their representations. Appendix B.4 lists detailed configurations.

**Metrics.** Scores use post-controller decisions; immediate and scheduled services count as INTER-**VENUE**. **Relevance** (precision) is the share of interventions matching reference opportunities; **Coverage** (recall) is the share of opportunities recovered. **Effectiveness** (F1), their harmonic mean, balances relevant interventions against missed opportunities. **False Positive Rate (FPR)**, also called False Intervention Rate (FIR), is the share of reference negatives receiving interventions. **Balance** (balanced accuracy) averages Coverage and the share of reference negatives correctly left without intervention. Higher is preferred except for FPR. **Intervention Rate (IR)** is the share of all events receiving interventions, without a preferred direction.

## 5.2 Main Results

ProactMem achieves the highest Effectiveness and Balance in every scope (Table 2, RQ1), exceeding the strongest external baseline’s Effectiveness by 4.3–15.7 percentage points.

Comparisons with external methods. Against A-MEM, ProactMem recovers more opportunities with fewer false interventions in every scope. However, Mem0 has higher Sports Coverage, and some methods attain lower Home/Code FPR with lower Coverage. IR combines recovered opportunities and false interventions; lower frequency alone does not imply better decisions.

**Reference sensitivity.** Rescoring fixed ProactMem outputs from a separate 450-target diagnostic against the earlier and separate-team references (Section 4.2) changes Effectiveness from 42.6%

to 42.5% on Sports, 68.7% to 55.6% on Home, and 57.6% to 47.4% on Code. Absolute scores are label-sensitive, especially on Home and Code.

**Episode-level analysis.** A model-assisted post-hoc analysis on those targets reconstructs 67 goal-and-object episodes from 76 positive event anchors. First timely matching assistance covers an episode; repetition without actionable new evidence is redundant. ProactMem covers 23/26 Sports episodes, 10/10 Home episodes, and 14/31 Code episodes. Its redundant-intervention shares are 1.2%, 43.6%, and 9.5%, respectively: full Home coverage co-exists with 17 redundant interventions out of 39, while Code leaves 17 reference episodes uncovered. Episode-level analysis complements event-level evaluation by measuring opportunity coverage and repeated interventions. Appendix D retains reference-sensitivity scores and episode-matching records.

## 5.3 Ablation Study and Memory Analysis

**Memory levels.** Event-only supplies current and bounded recent events; Event+Daily adds source-checked daily trajectories; ProactMem adds temporal patterns. These three variants in Figure 3 share Luna, decision prompt, schema, event bound, and controller. The controller uses representation-dependent keys, so the comparison measures memory-layer contributions within the implemented pipeline. Daily trajectories recover more opportunities, most on Code, with unchanged Sports/Home false-positive counts and two additional Code false positives. Patterns further increase Effectiveness by 9.9/13.5/7.0 points (Sports/Home/Code), increasing true positives and reducing false positives (RQ2). Daily abstraction expands context; supported, currently applicable patterns improve selectivity.

**Validation mechanisms.** Table 3 fixes model, prompt, event bound, context capacity, and controller. *w/o Replay Gate* bypasses activation thresholds for well-formed, source-linked, online-reconstructable Candidates, retaining current-applicability matching. *w/o Daily Validation* stores schema-valid, same-entity daily extractions without fact checking, retaining weekly construction and replay. Both removals lower Relevance and Effectiveness (RQ3). Replay gating contributes more Effectiveness and improves Coverage in every scope. On Code, daily validation removes 43

Table 2: Full-scope results (% , Luna). Relevance: share of interventions matching opportunities; Coverage: share of opportunities recovered; Effectiveness: combined Relevance–Coverage score (F1); FPR: share of reference negatives receiving interventions; Balance: average of Coverage and correct-silence rate; IR: share of events receiving interventions. Bold/underlining mark best/second-best directional scores per scope, including ties; IR has no preferred direction.

| Scope  | Method         | IR   | Relevance $\uparrow$ | Coverage $\uparrow$ | Effectiveness $\uparrow$ | FPR $\downarrow$ | Balance $\uparrow$ |
|--------|----------------|------|----------------------|---------------------|--------------------------|------------------|--------------------|
| Sports | ProactiveAgent | 69.0 | 30.0                 | <u>82.5</u>         | 43.9                     | 64.5             | <u>59.0</u>        |
|        | ContextAgent   | 54.3 | <u>31.0</u>          | 67.1                | 42.4                     | <u>50.0</u>      | 58.6               |
|        | Mem0           | 85.2 | 28.8                 | <b>97.8</b>         | <u>44.5</u>              | 81.0             | 58.4               |
|        | A-MEM          | 60.5 | 30.6                 | 74.0                | 43.3                     | 56.0             | <u>59.0</u>        |
|        | Graphiti       | 55.1 | 29.3                 | 64.4                | 40.3                     | 52.0             | 56.2               |
|        | ProactMem      | 53.8 | <b>35.8</b>          | 76.7                | <b>48.8</b>              | <b>46.1</b>      | <b>65.3</b>        |
| Home   | ProactiveAgent | 9.2  | 34.2                 | 26.8                | 30.1                     | <b>6.8</b>       | 60.0               |
|        | ContextAgent   | 16.1 | <u>35.9</u>          | 49.5                | 41.6                     | 11.7             | 68.9               |
|        | Mem0           | 8.5  | 23.9                 | 17.4                | 20.1                     | <u>7.3</u>       | 55.0               |
|        | A-MEM          | 19.0 | 34.6                 | <u>56.3</u>         | <u>42.9</u>              | 14.1             | <u>71.1</u>        |
|        | Graphiti       | 17.9 | 33.4                 | 51.1                | 40.4                     | 13.5             | 68.8               |
|        | ProactMem      | 17.7 | <b>48.6</b>          | <b>73.7</b>         | <b>58.6</b>              | 10.3             | <b>81.7</b>        |
| Code   | ProactiveAgent | 53.1 | 17.5                 | <u>78.2</u>         | 28.7                     | 49.7             | 64.2               |
|        | ContextAgent   | 32.0 | 22.5                 | 60.3                | 32.8                     | 28.2             | 66.1               |
|        | Mem0           | 7.7  | <u>39.8</u>          | 25.9                | 31.4                     | <b>5.3</b>       | 60.3               |
|        | A-MEM          | 30.4 | 30.4                 | 77.6                | <u>43.7</u>              | 24.0             | <u>76.8</u>        |
|        | Graphiti       | 28.1 | 26.8                 | 63.2                | 37.7                     | 23.3             | 69.9               |
|        | ProactMem      | 26.7 | <b>40.1</b>          | <b>89.7</b>         | <b>55.4</b>              | <u>18.1</u>      | <b>85.8</b>        |

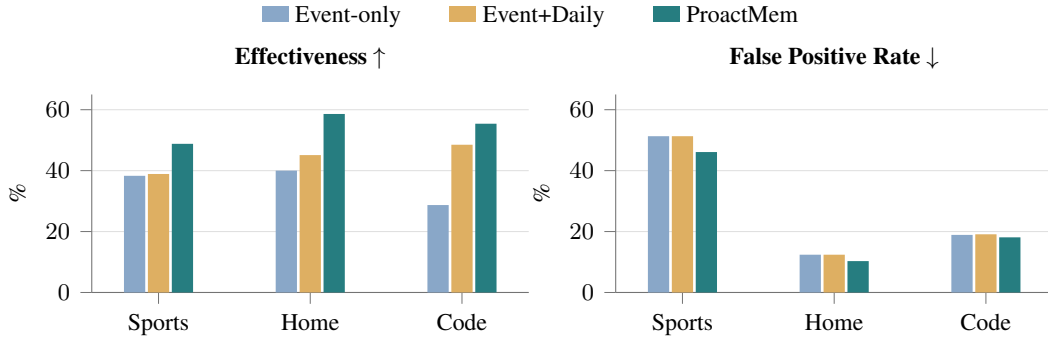


Figure 3: Memory-level ablation (% , Luna). Event-only, Event+Daily, and ProactMem share the decision prompt, schema, event bound, and controller policy, applied to each variant’s memory context. The panels report Effectiveness and FPR. Textual gains use unrounded scores.

false interventions without losing true positives, supporting improved selectivity.

**Episode control.** Before suppression scores recorded Code proposals with later states and decisions fixed, not a controller-free rerun. Filtering removes 66 false-positive and 17 true-positive proposals: Relevance and Effectiveness improve, but Coverage falls. Appendix C reports expanded ablation and controller scores.

## 5.4 Model Sensitivity

**Within-series pipeline sensitivity.** Table 4 jointly replaces decision and construction LLMs; benchmark, architecture, role-specific prompts, de-

coding, validation thresholds, and controller remain fixed. Luna–Terra–Sol improves Effectiveness and lowers FPR in every scope. This comparison measures pipeline sensitivity within one model series, with memory content and decision behavior changing together. Controlled ablations retain Luna.

**Cross-family Event-only comparison.** We compare GPT-5.6 Luna, Gemini 3.8 Flash, DeepSeek V4 Flash, and GLM 5.3 Flash. Task instructions, the 72-hour/100-event bound, and Episode Controller policy are shared. Figure 4 pools confusion counts across domains. Luna leads in F1 (37.4%) and recall; Gemini leads in precision.

Table 3: Validation ablations and Code controller diagnostic (% ,Luna). Higher is better. Before suppression scores recorded proposals with later states fixed. Bold/underlining mark best/second-best scores per column.

| Method               | Sports      |               | Home        |               | Code        |               |
|----------------------|-------------|---------------|-------------|---------------|-------------|---------------|
|                      | Relevance   | Effectiveness | Relevance   | Effectiveness | Relevance   | Effectiveness |
| w/o Replay Gate      | 31.1        | 42.7          | 43.6        | 54.4          | 33.8        | 48.1          |
| w/o Daily Validation | <u>34.4</u> | <u>47.5</u>   | <u>45.9</u> | <u>56.7</u>   | 36.1        | 51.5          |
| Before suppression   | —           | —             | —           | —             | <u>36.7</u> | <u>53.6</u>   |
| ProactMem            | <b>35.8</b> | <b>48.8</b>   | <b>48.6</b> | <b>58.6</b>   | <b>40.1</b> | <b>55.4</b>   |

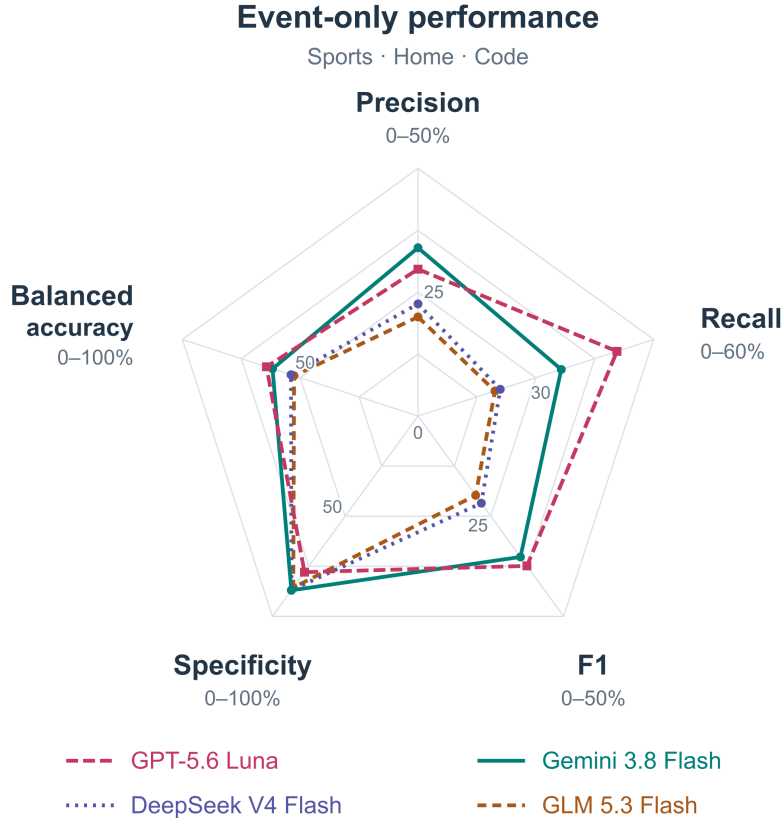


Figure 4: Single-run Event-only model comparison on three selected streams. Scores pool event-level confusion counts across domains; specificity equals  $1 - \text{FPR}$ . All axes start at zero, with metric-specific ranges.

## 6 Related Work

**Agent memory and long-horizon retrieval.** Generative Agents, MemoryBank, and MemGPT organize observations and experience through reflection, updating, or tiered context management (Park et al., 2023; Zhong et al., 2024; Packer et al., 2023). Mem0, A-MEM, and HippoRAG develop extracted memories, linked records, and graph-based retrieval (Chhikara et al., 2025; Xu et al., 2025; Gutiérrez et al., 2024). Graphiti maintains event-derived relations with temporal validity and provenance for retrieval (Zep, 2026). ProactMem derives recurring conditional patterns from daily trajectories, assesses their historical sup-

Table 4: Model sensitivity (%). Each cell gives Effectiveness  $\uparrow$  / FPR  $\downarrow$ . Bold/underlining mark best/second-best per scope. Construction and decision LLMs change together.

| Model | Sports                    | Home                     | Code                      |
|-------|---------------------------|--------------------------|---------------------------|
| Luna  | 48.8 / 46.1               | 58.6 / 10.3              | 55.4 / 18.1               |
| Terra | <u>54.1</u> / <u>39.4</u> | <u>64.4</u> / <u>9.1</u> | <u>60.3</u> / <u>14.9</u> |
| Sol   | <b>58.6</b> / <b>34.4</b> | <b>69.3</b> / <b>7.5</b> | <b>66.4</b> / <b>11.8</b> |

port, and matches applicable patterns to incoming events.

**Proactive decision and memory.** ProactiveAgent studies whether an agent should propose a task or remain silent and uses annotated event data to elicit this capability (Lu et al., 2025). ContextAgent combines sensory contexts with historical personas to anticipate service needs (Yang et al., 2025), while PASK couples streaming demand detection with hybrid long-term memory (Xie et al., 2026). Memory-oriented proactivity also supports internal reasoning and execution: MemCog integrates proactive memory exploration into conversational reasoning (Li et al., 2026), and Remember When It Matters uses a separate memory agent to inject selected reminders into long-horizon action agents (Wu et al., 2026). ProactMem focuses on constructing service-relevant temporal memory through source checking, historical replay, and current-event matching.

**Evaluation of proactive service.** LoCoMo and LongMemEval assess long-term conversational memory and reasoning (Maharana et al., 2024; Wu et al., 2025). Proactive benchmarks examine task proposals, latent-need detection, or memory triggering (Lu et al., 2025; Xie et al., 2026; Li et al., 2026). ProactiveStream evaluates sequential INTERVENE/ABSTAIN decisions as agents accumulate memory within isolated, causally replayed event streams, using episode-aware references to assess intervention timing. Research on notification timing further shows that interruption cost depends on task phase and receptivity (Adamczyk and Bailey, 2004; Iqbal and Bailey, 2007, 2008; Mehrotra et al., 2016). This motivates evaluating opportunity recovery together with unnecessary intervention.

## 7 Conclusion

We presented ProactMem, a proactive memory framework that combines bounded recent events, source-checked daily trajectories, and replay-assessed temporal patterns. Across the three scopes of ProactiveStream, ProactMem achieves higher event-level Effectiveness and Balance than the evaluated external baselines. Ablations support the contributions of hierarchical organization, daily fact validation, and historical replay gating. These findings highlight the value of matching source-grounded, historically supported memory to cur-

rent observations for service-opportunity detection.

## Limitations

ProactiveStream covers three low-risk service scopes and 17 streams under a protocol fixed after joint method and benchmark development. Independent held-out evaluation and cross-family model transfer remain future work. Reference labels represent service opportunities under the annotation protocol; the 450-target review assesses three selected streams. The Code suppression diagnostic holds later states fixed; controller-free behavior requires a sequential comparison. Historical replay assesses observed support; prospective validity under changing behavior requires further study. Our offline metrics evaluate opportunity detection. Recipient benefit, interruption burden, end-to-end cost, and deployment safety require prospective measurements. The reported results support comparisons within the evaluated protocols, with broader generalization depending on additional populations, observation conditions, and model families.

## Ethical Considerations

Our evaluation is limited to optional activity support, gentle check-ins, and low-risk repository assistance. Medical and high-risk safety decisions require separate domain-specific validation and safeguards. Inference is isolated by stream, and reference labels, episode annotations, and scoring metadata are excluded from model inputs.

Longitudinal events and derived memories can reveal sensitive routines even when identifiers are pseudonymized. Event content, prediction traces, and inferred trajectories require access and retention protections; external-model use should minimize transmitted information and respect source-data permissions.

Deployment should require informed opt-in and allow inspection, correction, deletion, retention control, and disabling proactive assistance. An intervention prediction represents a service proposal; consequential action requires separate authorization. These deployment safeguards require separate implementation and evaluation.

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## A Benchmark and Annotation Protocol

### A.1 Public Events and Availability

**Sports.** HeartSteps mobile activity contexts (Klasnja et al., 2019) supply 1,456 meta-events from nine complete participant streams over 367 observed participant-days. Public fields include decision-time availability, recognized activity, recent steps, location category, weather, and local decision slot. Randomized treatment assignments and future behavioral outcomes are excluded; services concern optional activity support.

**Home.** CASAS ambient-sensor traces (Cook et al., 2013) supply four homes with 14 consecutive days each. Per local day, ten-minute source blocks are assigned exactly once to 28 regular 50-minute segments and one 40-minute tail, yielding 406 events per home. A segment becomes available only after interval closure and contains within-interval motion counts, room sequences, door actions, and compact numeric summaries. Gentle check-ins exclude disease, fall, travel, danger, and unsupported preference inference.

**Code.** Public repository activity, including GH Archive records (Grigori, 2026), supplies 1,459 events across four repositories with 174 reference interventions. Pushes, issue updates, pull-request actions, and reviews retain GitHub’s native event granularity, with one decision per event. Records are ordered by timestamp and then event ID. Each repository owns isolated memory and intervention state even when contributors overlap. Services concern low-risk triage, review, and maintenance; a native PushEvent is an observation, not an agent intervention.

**Public-input contract.** Both label classes share the same scope-specific schema: event and stream identifiers, timestamps, and contemporaneously observable fields. Reference labels, candidate flags, episode annotations, memory-level markers, and scoring masks remain evaluator-private. Evaluation includes early events before daily trajectories or temporal patterns are available.

### A.2 Reference Labels and Causal Replay

Annotators apply the opportunity and novelty semantics in Section 4.2 to each current event and its complete prior same-stream history, independently of a tested method’s memory. LLM-assisted

screening prioritizes the review queue; three team members independently assign labels while blinded to evaluated-system decisions, followed by cross-review and adjudication of every public event.

The Home aggregation rule precedes event construction. Each evaluation version records the event inventory, labels, intervention prompt, Event Memory bound, and validation settings; these are frozen before the corresponding main runs. Evaluation follows a fixed protocol on the jointly developed benchmark.

Every stream starts from empty method state and is replayed chronologically under the causal execution order in Section 3.3. Current events are committed only after the output is finalized; maintenance incorporating an event affects later decisions. Labels are joined offline, and reference episode states and scoring metadata remain organizer-private.

## B Implementation and Experimental Configuration

### B.1 Daily Evidence and Replay

Daily construction uses an independent buffer of all committed events from the open local day, cleared after processing. Activity segments contain a local interval, typed observed, context-before, and outcome fields, plus source-event identifiers. The checks in Equation 5 canonicalize scalar values, accept exact or schema-preserving suffix field paths, and require consistent entities, intervals, and resolved citations. Validation and replay operate on typed evidence. Accepted trajectories remain in the current construction cycle after their events leave Event Memory.

A cycle contains seven completed daily records, each assigned to one cycle, and may span more than seven calendar days. Empty or unobservable records also advance the cycle. Consolidation resets the online daily view while preserving records for audit. Replay follows Equation 6: a triggered, observable day is fulfilled when the expected pattern occurs at or after the trigger within the window; otherwise it is violated. Missing observations are censored, each date counts at most once, and the base rate includes triggered and other observable days under the same time and weekday restrictions. Confidence is zero with no eligible days; lift is zero with zero base rate.

## B.2 Lifecycle and Current Matching

Lifecycle eligibility is separate from occurrence state and service-episode state. After admission, confidence combines replay counts and subsequent outcomes in a support ratio smoothed with one initial success and one initial failure; censored observations preserve both counts. Two consecutive violations move an Active pattern to *weakening* with lower retrieval priority; four consecutive violations move it to *dormant*. A fulfilled observation restores a weakening pattern to Active when confidence is at least 0.5. Dormant patterns must return to Candidate and pass replay readmission.

**Current matching.** Current matching is deterministic. Retrieval admits upcoming, pending, or deviated occurrences of Active or weakening patterns. Occurrence priority is deviated, followed by pending and upcoming.

## B.3 Episode Identity

In Sports/Home, the intervention key combines scope-specific episode state, proposed service action and target, and sorted pattern identifiers available in context. Plain summaries or graph facts supply an empty pattern-identifier list. Sports state includes availability/activity; Home state includes activity/room and onset, confirmed, and sustained phases at one, two, and three or more consecutive matching observations. Without an explicit episode key, scope-specific time or target fields identify the opportunity, with unchanged evidence additionally required for duplicate suppression. State changes, sequence restarts, service or pattern identifier changes, and local-day transitions can yield new keys.

Code uses repository-object identity and the most recent allowed intervention’s mode, recipient, content, and firing condition. An exact match converts a repeated create proposal to ABSTAIN; changed signatures and other episode operations pass this check. Representation-dependent key evidence can yield different suppression decisions across memory interfaces.

## B.4 Model and Method Interfaces

Decision and memory-construction calls use the model, API, and decoding settings in Section 5.1, with separate output budgets. Decision limits are interface-specific (256 or 512 tokens); construction limits are configured by call role. Table 5 consolidates the external-method interfaces. Methods

retain their own prompts, retrieval, and computational budgets.

Graphiti supplies relation facts directly, using the task policy and output contract with memory-specific guidance; its ingestion episodes correspond to source records. The cross-family Event-only comparison uses backend-specific output-format and recovery settings.

**Scoring.** Using final-decision confusion counts, Relevance is  $TP/(TP + FP)$ , Coverage is  $TP/(TP + FN)$ , Effectiveness is  $2TP/(2TP + FP + FN)$ , and Balance is  $\frac{1}{2}[TP/(TP + FN) + TN/(TN + FP)]$ .

Table 5: External-method memory interfaces and implementation settings.

| Method         | Interface and settings                                                                                                                                                                                                                                                                                                  |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ProactiveAgent | Released evaluation prompt and proposal/null contract, adapted to the study backbone; causal event–decision history capped at 1,000 events including the current event per request (Lu et al., 2025; THUNLP, 2026).                                                                                                     |
| ContextAgent   | In-context proactive-service formulation adapted to public event representations and the study backbone (Yang et al., 2025).                                                                                                                                                                                            |
| Mem0           | Open-source mem0ai==2.0.12 extraction, consolidation, and retrieval; local all-MiniLM-L6-v2 embeddings; top-8 same-entity retrieval with search threshold 0.1 (Chhikara et al., 2025; Mem0, 2026).                                                                                                                      |
| A-MEM          | Linked notes with association updates; retrieved notes augment the bounded event context (Xu et al., 2025).                                                                                                                                                                                                             |
| Graphiti       | Graphiti 0.30.2 with Neo4j and local 384-dimensional all-MiniLM-L6-v2 embeddings; committed events enter a stream-isolated graph. The current event queries native hybrid edge search with reciprocal-rank fusion, returning at most ten relation facts with validity fields and source-record identifiers (Zep, 2026). |

## C Additional Ablation and Controller Results

Table 6 supplements the Relevance and Effectiveness scores in Table 3 with intervention frequency and Coverage. On Code, suppression removes 66 false-positive and 17 true-positive proposals; Coverage falls from 99.4% to 89.7%. This scores recorded proposals with later states fixed, not a controller-free sequential rerun. For the memory hierarchy in Figure 3, the aggregate net TP increments are +4/ +15/ +64 from daily trajectories and +57/ +33/ +22 from temporal patterns (Sports/Home/Code); they are net differences, not event-level causal attributions.

## D Reference and Episode Diagnostics

### D.1 Reference Sensitivity

The diagnostic selects 150 consecutive targets from one stream per scope using a fixed SHA256 ranking of eligible streams and contiguous starts based only on public metadata, with at least seven days of preceding history. The streams are Sports participant\_35, Home home\_0004, and Code repository callstack/react-native-pager-view. Their target periods are December 16, 2015–January 18, 2016; June 23–28, 2011; and June 21–November 4, 2025. All 37, 203, and 152 preceding events are processed chronologically, respectively: 842 public events supply context for 450 scored targets.

A separate three-reviewer team independently labels each target with its prior same-stream history while blinded to evaluated-system outputs, then discusses judgments to obtain consensus references. Relative to the earlier diagnostic references, Sports/Home have 13/13 changes from IN-

Table 6: Additional evidence-check ablation and Code controller metrics (%). IR has no preferred direction; higher Coverage is preferred. Bold/underlining mark best/second-best Coverage per scope, including ties. Relevance and Effectiveness appear in Table 3.

| Scope  | Method               | IR   | Coverage $\uparrow$ |
|--------|----------------------|------|---------------------|
| Sports | w/o Replay Gate      | 55.0 | <u>68.2</u>         |
|        | w/o Daily Validation | 55.8 | <b>76.7</b>         |
|        | ProactMem            | 53.8 | <b>76.7</b>         |
| Home   | w/o Replay Gate      | 19.3 | 72.1                |
|        | w/o Daily Validation | 18.9 | <b>74.2</b>         |
|        | ProactMem            | 17.7 | <u>73.7</u>         |
| Code   | w/o Replay Gate      | 29.4 | 83.3                |
|        | w/o Daily Validation | 29.6 | <u>89.7</u>         |
|        | Before suppression   | 32.4 | <b>99.4</b>         |
|        | ProactMem            | 26.7 | <u>89.7</u>         |

TERVENE to ABSTAIN and 4/0 changes in the reverse direction; Code has 6 removals and 16 additions. These counts describe differences between the earlier diagnostic references and the separate team’s judgments on 450 targets across three selected streams.

Table 7 scores identical recorded ProactMem outputs against these two sets of references. The predictions come from separate diagnostic runs on the selected targets. The diagnostic uses Luna with temperature zero, thinking disabled, decision limits of 256 tokens for Sports/Home and 512 for Code, and the same 72-hour/100-event bound, daily validation, replay gating, and episode-control policy. Immediate and scheduled services both count as INTERVENE.

Table 7: Reference sensitivity on 150 targets per scope.  $I$  is the number of final interventions; positives and confusion counts use the separate team’s references. Effectiveness is F1 (%). Earlier diagnostic references use original Sports/Home labels and 122 original plus 28 additional Code annotations. Both columns score the same diagnostic outputs.

| Scope  | Positives | $I$ | TP | FP | FN | TN  | Earlier diagnostic Effectiveness | Separate-team Effectiveness |
|--------|-----------|-----|----|----|----|-----|----------------------------------|-----------------------------|
| Sports | 27        | 86  | 24 | 62 | 3  | 61  | 42.6                             | 42.5                        |
| Home   | 15        | 39  | 15 | 24 | 0  | 111 | 68.7                             | 55.6                        |
| Code   | 34        | 42  | 18 | 24 | 16 | 92  | 57.6                             | 47.4                        |

Table 8: ProactMem episode diagnostic on 150 targets per scope. Covered/New includes Episode Coverage (%) in parentheses; SR and RIR are percentages.  $I$  is the sum of first valid ( $F$ ), valuable follow-up ( $V$ ), redundant ( $R$ ), uncertain ( $U$ ), unmatched, late, and unresolved outcomes.  $U = 0$  in all three scopes; unresolved semantic or condition matches remain separate from supported or redundant assistance.

| Scope  | Covered/New (Coverage) | $I$ | $F$ | $V$ | $R$ | Unmatched | Late | Unresolved | SR   | RIR  |
|--------|------------------------|-----|-----|-----|-----|-----------|------|------------|------|------|
| Sports | 23/26 (88.5)           | 86  | 23  | 1   | 1   | 60        | 0    | 1          | 27.9 | 1.2  |
| Home   | 10/10 (100.0)          | 39  | 10  | 6   | 17  | 6         | 0    | 0          | 41.0 | 43.6 |
| Code   | 14/31 (45.2)           | 42  | 14  | 3   | 4   | 16        | 4    | 1          | 40.5 | 9.5  |

## D.2 Episode References and Matching

Each reference specifies a service goal and object, causal evidence, onset, explicitly listed valid observations, and any deadline or closure evidence. Semantic matching uses these fields to associate interventions with reference episodes.

**Matching and follow-up.** Coverage follows ProactMem’s own interventions, initialized from recorded prefix outputs; fixed event-label novelty instead follows the reference sequence (Equation 9). Only target interventions are scored. An intervention must match the goal and object at a listed valid observation. Its proposed service time is the decision time plus its specified delay (zero for immediate service), and must meet any explicit episode deadline. The first timely match covers the episode. Each output receives at most one credit: the first uncovered compatible episode in the fixed reference order, otherwise the first already covered one.

Subsequent matches are compared with the method’s preceding assistance for that episode. A valuable follow-up requires new causal evidence that changes the actionable service or next step, and a message addressing that change. Rephrasing or continued status alone is redundant; value requiring unobserved delivery, recipient response, or completion remains uncertain. Valuable follow-ups do not create additional covered episodes.

**Metrics and boundaries.** Let  $I$  count all target interventions,  $F$  first valid interventions,  $V$  valuable follow-ups,  $R$  redundant follow-ups, and  $U$  uncertain follow-ups. Episode Coverage divides covered new episodes by all new reference episodes; Supported Relevance is  $SR = (F + V)/I$ , and Redundant Intervention Rate is  $RIR = R/I$ . Uncertain, unmatched, late, and unresolved outputs remain in  $I$  but receive no supported-assistance credit. Table 8 combines the rates and outcome counts.

Coverage is conditional on the observed segment and onset convention. The main diagnostic treats the first Home opportunity as target-onset; treating it as a continuation from the prefix instead gives 9/9 new opportunities covered. These exploratory scores depend on the constructed references and semantic matches.